VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering

Project Report on

InterviewIQ – An AI-Powered Mock Interview Platform

In partial fulfillment of the Fourth Year (Semester–VII), Bachelor of Engineering (B.E.) Degree in Computer Engineering at the University of Mumbai

Academic Year 2025-26

Project Mentor - **Prof.Vidya S. Zope**

Submitted by

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(2025-26)

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CERTIFICATE

This is to certify that **Toshit Hole(D17C-28),Shreyas Bagwe(D17C-05),Jatin Hargunani(D17C-27),Subodh Thalkari(D17C-62)**of Fourth Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on "**Interview-IQ:An AI Mock Interview Platform**" as a part of the coursework of PROJECT-I for Semester-VII under the guidance of **Prof.Vidya Zope** in the year 2024-2025.

Date

Internal Examiner

External Examiner

Project Mentor

Head of the Department
Dr. Mrs. Nupur Giri

Principal
Dr. J. M. Nair

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea / data / fact / source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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We express our hearty thanks to them for their assistance without which it would have been difficult in finishing this project synopsis and project review successfully.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	**Description of the Course Outcome**
CO 1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamentals in the domain of practical applications for problem identification, formulation and solution
CO 3	Attempt & Design a problem solution in the right approach to complex problems
CO 4	Cultivate the habit of working in a team
CO 5	Correlate the theoretical and experimental/simulation results and draw the proper inferences
CO 6	Demonstrate the knowledge, skills and attitudes of a professional engineer, & prepare a report as per the standard guidelines.

Abstract

InterviewIQ – An AI-Powered Mock Interview Platform is an advanced and intelligent system designed to revolutionize the way individuals prepare for interviews by integrating artificial intelligence, natural language processing, and multimodal analysis. The platform leverages multiple Large Language Models (LLMs) to simulate highly realistic and adaptive interview experiences, providing users with an environment that mirrors real-world professional interactions. It offers both audio and video interview modes, where the audio module focuses on evaluating verbal attributes such as tone, fluency, filler word usage, long pauses, and overall speech confidence, while the video module—powered by OpenCV and MediaPipe—analyzes non-verbal communication aspects such as eye contact, gestures, facial expressions, and posture. A unique feature of InterviewIQ is its AI Interviewer Personalization System, allowing users to choose from four distinct virtual interviewer personas, each with different temperaments and questioning approaches, thereby offering a psychologically diverse and engaging experience. The platform is designed to provide in-depth analytics and performance feedback post-interview, including verbal and behavioral insights, confidence scores, and detailed evaluations highlighting strengths and areas for improvement. Furthermore, InterviewIQ integrates real-time socket-based communication for seamless interaction, low-latency response generation, and advanced speech-to-text synthesis for accurate evaluation. It also plans to introduce a Retrieval-Augmented Generation (RAG)-based feature, enabling users to upload personal documents or notes from which the AI interviewer can generate context-specific questions and assess responses accordingly. By combining the capabilities of multi-agent orchestration, cross-LLM intelligence, and adaptive multimodal evaluation, InterviewIQ offers a highly interactive, data-driven, and user-centric approach to interview preparation, empowering individuals to enhance their communication, confidence, and analytical skills, while bridging the gap between traditional mock interviews and next-generation AI-driven career readiness solutions.

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Chapter 1: Introduction

1.1 Introduction

Interview preparation plays a pivotal role in shaping an individual's professional journey, as it directly impacts employability, communication confidence, and overall career readiness. Despite the abundance of online learning resources, most candidates still struggle to find platforms that offer personalized, realistic, and adaptive practice environments. Conventional approaches—such as question banks, video tutorials, or manual mock sessions—fail to capture the dynamic and interactive nature of real interviews. These methods often lack flexibility, are costly, and provide minimal feedback on both verbal and non-verbal performance parameters.

With the rapid progress in Artificial Intelligence (AI) and Machine Learning (ML), intelligent interview simulation has emerged as a transformative solution for bridging this gap. AI-driven systems can simulate human-like interactions, generate adaptive questions, and evaluate candidate performance using speech, facial, and behavioral cues. However, most existing tools remain limited to text-based interfaces, offering little real-time adaptability or multimodal analysis. They often fail to assess communication confidence, body language, or engagement, which are crucial for holistic interview preparation.

InterviewIQ addresses these limitations by introducing a multi-agent, multimodal AI-powered mock interview platform that enables users to practice interviews through both audio and video modes. The system utilizes powerful LLM-based intelligence to create realistic, domain-specific, and context-aware interview experiences. It further evaluates users on speech patterns, filler words, gesture tracking, and confidence levels—providing detailed analytical feedback and performance reports. By combining adaptability, inclusivity, and interactivity, InterviewIQ aims to redefine how candidates prepare for real-world interviews.

1.2 Motivation

In today's competitive job landscape, candidates face immense pressure during interviews, often due to lack of realistic practice and constructive feedback. While professional coaching and paid mock interviews exist, they are expensive, geographically restricted, and non-scalable. Furthermore, the majority of existing online tools are limited to text-based question-answer modules, offering no real-time adaptability or behavioral assessment.

The motivation behind developing InterviewIQ stems from the need to democratize high-quality interview preparation through AI-driven intelligence. The system not only helps users improve their technical and communication skills but also focuses on confidence building by simulating real-life interaction scenarios. It offers multiple interview types such as Technical, HR, Finance, Marketing, and Sales, with subcategories like *Data Structures and Algorithms, Frontend, Backend, and DevOps* for technical interviews. This allows users to prepare for specific domains of their choice.

A distinctive motivation is to create diverse AI interviewer personas, enabling users to choose their interviewer's gender, tone, and attitude. The system includes four interviewer profiles—two female (a strict senior interviewer and a confident young professional) and two male (an experienced elder mentor and a young supportive interviewer). These personalities help simulate varied interview environments, reducing anxiety and improving adaptability. Thus, InterviewIQ's motivation lies in creating a personalized, affordable, and psychologically enriching learning experience for every user.

1.3 Problem Definition

Despite the technological progress in AI-based assessment tools, interview simulation remains underdeveloped in terms of realism, personalization, and feedback. Existing systems suffer from the following limitations:

- **Lack of Adaptability:** Most platforms use pre-defined question sets without context-aware follow-ups.
- **Absence of Multimodal Analysis:** Current solutions rarely combine audio and video cues for holistic evaluation.
- **Limited Diversity:** Users cannot choose interviewer behavior or tone, which is crucial for real-world readiness.
- **Inadequate Feedback:** Feedback is often generic and lacks detailed analytics on communication or behavioral traits.
- **Cost and Accessibility Issues:** Many tools rely on premium APIs or closed-source platforms, restricting access.

Hence, the core problem addressed by this project is:

To design and develop a scalable, multimodal AI-powered interview simulation platform that integrates multiple LLMs and computer vision techniques to evaluate users comprehensively across technical, communication, and behavioral parameters, while offering interviewer personalization and detailed analytical feedback.

1.4 Relevance of the Project

InterviewIQ holds significant relevance in the modern employment and education ecosystem, where virtual hiring and remote assessments are rapidly becoming the norm. Organizations are increasingly adopting AI-assisted recruitment processes, and candidates must adapt to these evolving interview environments. By providing a realistic, adaptive, and feedback-oriented mock interview platform, InterviewIQ aligns with global trends in AI-based education and professional training.

From an academic and technological perspective, the project contributes to the domains of Conversational AI, Computer Vision, and Speech Processing, offering practical insights into their integration. It supports the goals of AI-driven skill development and digital employability enhancement. Furthermore, the project aligns with the United Nations Sustainable Development Goal (SDG) 8 – Decent Work and Economic Growth, by enabling equitable access to career-preparation resources.

In summary, InterviewIQ's relevance is defined by its potential to:

- Enhance confidence and communication skills through realistic simulation.
- Bridge the accessibility gap by offering an affordable, cloud-based platform
- Enable fair and transparent evaluation using data-driven analytics
- Advance AI's role in ethical, inclusive, and personalized education.

1.5 Methodology Used

The development methodology for **InterviewIQ** is structured into distinct but interconnected stages that ensure accuracy, adaptability, and performance efficiency:

Step 1: Data and Model Integration

The platform integrates multiple LLMs (OpenAI, Claude, Gemini, Groq) for diverse question generation, response evaluation, and contextual adaptation. It leverages publicly available question datasets and fine-tuned dialogue prompts to ensure domain variety and natural flow.

Step 2: Multimodal Architecture Development

Two interaction modes—Audio and Video—are implemented. The Audio module utilizes Whisper ASR for speech-to-text conversion and Coqui TTS for AI interviewer voice synthesis. The Video module employs OpenCV and MediaPipe to detect facial expressions, eye contact, and gestures. Combined, these create a holistic feedback mechanism covering both verbal and non-verbal communication.

Step 3: Interviewer Personalization

A multi-agent architecture assigns unique behavioral parameters to each AI interviewer persona.

This affects question tone, difficulty, and interaction style, allowing users to experience varied psychological environments similar to real interviews.

Step 4: Analytics and Feedback System

Post-interview, the platform generates a comprehensive analytical report summarizing metrics like filler word frequency, pause duration, confidence score, facial engagement, and response relevance. These are visualized using dashboards for clarity and progress tracking.

Step 5: Deployment and Testing

The entire system is built using a MERN-based stack (MongoDB, Express, React, Node.js) with Python microservices for AI modules. The web application will be deployed on cloud infrastructure, ensuring accessibility and scalability. User feedback and continuous testing will guide refinements for Stage-2 development.

Chapter 2: Literature Survey

2.1 Research Papers Referred

Sr no	Paper Title & Authors	Year	Description / Methodology	Inference / Relevance to Project
1	Chen, L., Wang, S., & Liu, H. – “AI in Education: Applications and Methodologies”[1]	2023	Discusses the application of AI models in educational systems for adaptive learning and student evaluation.	Establishes how AI can personalize assessment processes and improve learner outcomes.
2	Orak, C. & Turan, Z. – “Using AI in Digital Interview and Assessment Systems”[10]	2024	Explores the use of NLP and sentiment analysis in virtual interviews to measure communication skills.	Provides insight into using NLP for candidate evaluation, applicable to InterviewIQ’s feedback module.
3	Patel, D., & Mehta, V. – “Speech Recognition for Interactive Learning Environments”[11]	2023	Describes speech recognition (ASR) integration for interactive educational tools.	Supports InterviewIQ’s use of Whisper ASR for speech-to-text conversion.
4	Gupta, P., Sharma, K., & Rao, A. – “Machine Learning Models for Soft Skill Assessment”[9]	2022	Describes speech recognition (ASR) integration for interactive educational tools.	Reinforces the need for analytics-driven evaluation of communication and personality traits.
5	Krishnan, N. – “AI Agents: Architecture and Real-World Applications”[8]	2025	Explains how multi-agent systems coordinate tasks in complex AI applications.	Provides a foundation for InterviewIQ’s multi-agent orchestration design.

2.2 Patent Search Summary

Sr.No	Patent Title	Patent No. / Year	Description / Methodology	Relevance to Interview-IQ
1	System and Method for Conducting Virtual Interviews Using	US 20210156789 A1 (2021)	Describes an AI-driven virtual interview system that analyzes user responses and facial expressions to evaluate	Provides foundational insight into integrating AI for candidate evaluation and emotion recognition during

	Artificial Intelligence		suitability for a role. The patent focuses on automated question selection and sentiment-based scoring.	interviews.
2	Interactive Interview Simulation Platform with Speech and Emotion Recognition	EP 3856721 A2 (2022)	Outlines a system capable of conducting voice-based interviews where candidate responses are transcribed and analyzed for tone, confidence, and emotion. Uses ASR and NLP for adaptive questioning.	Supports the development of InterviewIQ's audio-interview mode using speech analytics and filler word detection.
3	Method for Generating Dynamic Question Sets in AI-Based Training Systems	US 20220091457 A1 (2022)	Focuses on dynamically generating question sets using machine learning models based on candidate profiles and performance history.	Aligns with InterviewIQ's adaptive question-generation system powered by LLMs for context-aware questioning.
4	Computer Vision-Based Behavioral Assessment System	WO 2020034781 A1 (2020)	Utilizes computer vision techniques to monitor eye gaze, facial expressions, and body gestures during digital interviews. Scores candidates on non-verbal engagement and confidence.	Forms the conceptual basis for InterviewIQ's video interview mode using OpenCV and MediaPipe for gesture and eye-contact analysis.

2.3 Existing Systems

Sr no	Tool Name	Key Features	Target Users	Limitations
1	Final Round AI	Provides real-time AI assistance during live interviews; supports answer structuring, coding help, and JD-based preparation.	Job seekers needing real-time guidance during technical or HR interviews.	May raise ethical concerns if used during actual interviews; limited learning feedback.
2	Remasto AI	Offers mock interviews based on job roles or companies; includes resume builder and basic feedback analytics.	Students and professionals preparing for role-specific interviews.	Text-based only; lacks voice interaction and adaptive learning capability.

3	Interviews Chat	Conducts AI-based mock interviews with smart follow-up questions and performance summaries; includes resume and cover letter review.	Candidates preparing for behavioral and technical interviews.	No voice or video-based interaction; feedback is generic and not deeply personalized.
4	Sapia AI	Enterprise tool for AI-driven candidate screening; performs text-based, bias-free assessments with personality analytics.	HR teams and recruiters conducting large-scale hiring.	Designed for companies, not learners; lacks real-time or speech-based interaction.

2.4 Lacuna in Existing Systems

While existing systems demonstrate the growing influence of AI in interview preparation and recruitment, several gaps remain unaddressed:

1. **Absence of Speech Interaction:** Most tools rely solely on text-based inputs without incorporating natural speech recognition or voice synthesis.
2. **Lack of Real-Time Adaptability:** Questions are pre-defined and do not evolve based on user responses.
3. **Limited Educational Feedback:** Current platforms focus on evaluation rather than continuous learning improvement.
4. **High Cost or Restricted Access:** Some systems, such as Sapia AI, are enterprise-only and not freely accessible to individual learners.

2.5 Comparison of Existing Systems and Proposed Work

Sr no	Existing Systems	Proposed Work
1	Existing platforms like Final Round AI, Remasto AI, Interviews Chat, and Sapia AI primarily offer text-based or limited interactive mock interview experiences. These systems generate pre-defined questions without real-time adaptability and often lack multimodal input support.	InterviewIQ provides fully interactive audio and video-based interviews powered by LLMs capable of adapting questions dynamically to user responses, ensuring a realistic and personalized experience.
2	Most existing tools focus only on textual responses and ignore non-verbal communication elements such as tone, pauses, or gestures.	InterviewIQ integrates speech analysis (detecting filler words, pauses, and tone) and computer vision analytics (eye contact, facial expressions, gestures) for holistic evaluation.

3	Feedback provided in existing systems is generic or score-based, offering limited actionable insights.	InterviewIQ generates comprehensive performance reports with analytics on communication clarity, confidence, domain knowledge, and behavioral traits.
4	Existing solutions do not allow interviewer personality customization and treat every interaction the same way.	InterviewIQ introduces AI interviewer personas—four distinct profiles (two male, two female) with varying tones and strictness, allowing users to choose interviewer preferences.

2.6 Focus Area

The primary focus area of InterviewIQ is to develop a multimodal, adaptive, and intelligent interview simulation platform that provides a realistic and personalized experience to every user. The system emphasizes AI-driven question adaptability, speech and gesture-based evaluation, and customizable interviewer personas to simulate diverse real-world interview environments. By integrating advanced LLM-based reasoning, speech analytics, and computer vision modules, InterviewIQ ensures holistic assessment across technical, communication, and behavioral dimensions. The project's central aim is to make interview preparation interactive, data-driven, and accessible, enabling users to gain confidence, receive constructive insights, and enhance their employability through continuous AI-guided improvement.

Chapter 3: Requirements

3.1 Proposed model

The proposed model of InterviewIQ follows a modular, service-oriented architecture designed for scalability, adaptability, and real-time performance. It is structured into three core layers — the Presentation Layer, Application Layer, and Data & AI Layer. The Presentation Layer, built using React.js and Tailwind CSS, serves as the user interface where candidates interact through audio or video interviews. The Application Layer, developed with Node.js and Express, manages communication between the frontend and backend through secure RESTful APIs and WebSockets for real-time data flow.

The AI and Data Layer integrates multiple components, including LLM-based question generation, speech analysis (Whisper ASR), and computer vision modules (OpenCV & MediaPipe). These components collectively handle natural language understanding, filler word detection, facial and gesture analysis, and feedback synthesis. All data—such as user profiles, session records, and analytics reports—are securely stored in MongoDB. This layered architecture ensures modularity, allowing each subsystem (AI, video processing, analytics, and storage) to function independently while maintaining seamless integration across the platform.

3.2 Functional Requirements

The functional requirements of InterviewIQ define the core operations and behaviors expected from the system to ensure smooth and intelligent interview simulation.

Key Functional Requirements:

1. **User Registration and Authentication** – The system must allow users to securely register, log in, and manage profiles.
2. **Interview Mode Selection** – Users should be able to choose between audio and video interview modes.
3. **Domain and Subdomain Selection** – Users must select the interview category (e.g., Technical, Finance, Marketing, Sales) and subdomain (e.g., DSA, Frontend, Backend).
4. **AI Interviewer Personalization** – The system should enable users to select from four AI interviewer personas (strict/lenient, male/female).

5. **Dynamic Question Generation** – The AI module should generate context-aware, adaptive questions using LLMs based on user responses.
6. **Speech Processing** – In audio interviews, the system must analyze speech for filler words, long pauses, tone, and fluency.
7. **Video Analysis** – In video interviews, the system should use OpenCV and MediaPipe to evaluate eye contact, gestures, and facial expressions.
8. **Real-time Feedback** – The system should provide real-time conversational flow between the AI interviewer and the user.
9. **Performance Analytics Generation** – After each interview, the system must generate a detailed report showing scores for technical accuracy, communication, confidence, and behavioral indicators.
10. **Interview History Management** – The platform must store all past interview sessions and analytics for future reference.
11. **Admin Dashboard** – An admin module should monitor usage statistics, feedback trends, and system performance.
12. **Cloud Deployment and Accessibility** – The system should be deployable on cloud servers, allowing access from any device through a web browser.

3.3.Non-Functional Requirements

- **Performance:** Real-time AI processing must be efficient, though slight latency may occur.
- **Usability (UX):** Must feature a user-friendly interface.
- **Reliability:** The system must ensure accurate speech recognition (Whisper ASR) and realistic voice synthesis (Coqui TTS).
- **Scalability:** Must be able to handle increasing user traffic and data.
- **Security:** Implement authentication middleware and API request validation for secure operations.

3.4 Hardware & Software Requirements

The development and deployment of InterviewIQ require a combination of suitable hardware infrastructure and robust software tools to ensure smooth operation, scalability, and efficient performance.

Hardware Requirements:

- **Development System:** Laptop/PC with minimum Intel i5 / AMD Ryzen 5 processor or higher.
- **Memory (RAM):** Minimum 8 GB RAM (16 GB recommended for smoother AI and video processing).
- **Storage:** Minimum 256 GB SSD for local development and dataset storage.
- **Graphics Support:** GPU-enabled system (optional but beneficial for faster AI inference).
- **Internet Connectivity:** Stable broadband connection for accessing cloud services and APIs.
- **Deployment Server:** Cloud-based virtual machine (AWS EC2 / Google Cloud / Render) with 4 vCPUs, 8 GB RAM, and Docker support.

Software Requirements:

- **Operating System:** Windows 10/11, macOS, or Ubuntu Linux.
- **Frontend:** React.js, Tailwind CSS.
- **Backend:** Node.js, Express.js.
- **Database:** MongoDB (v5.0 or later).
- **AI and Machine Learning Modules:** Python (v3.10+), LangChain, Whisper ASR, Coqui TTS, OpenCV, MediaPipe.
- **API and Development Tools:**
    - **Postman** – for testing and debugging backend APIs.
    - **Docker & Docker Compose** – for containerized deployment of microservices.
    - **Git & GitHub** – for version control and collaboration.
    - **VS Code** – for code development and debugging.
- **Web Communication Frameworks:** WebRTC, Socket.io for real-time interview interactions.
- **Cloud Deployment Tools:** NGINX (reverse proxy and static file serving), Render / Vercel for hosting the application.

This configuration ensures that InterviewIQ runs efficiently across development, testing, and deployment environments, providing a stable and scalable infrastructure for multimodal AI-driven interviews.

3.5.Constraints of working System(Interview-IQ)

- **API Dependency:** The system's performance depends on third-party AI APIs for language, speech, and vision processing, which may introduce latency or downtime.
- **High Computational Demand:** Real-time audio and video processing, along with LLM inference, require substantial processing power and memory resources.
- **Network Reliability:** The platform's real-time interaction relies heavily on stable internet connectivity; poor bandwidth can disrupt interviews or delay responses.
- **Operational Cost:** Using premium APIs and cloud-based GPU instances increases development and maintenance costs.
- **Data Privacy and Security:** Storing and processing user audio/video data must comply with strict security standards to ensure confidentiality and prevent misuse.
- **Limited Time for Development:** Academic project timelines restrict the integration of advanced features and extensive user testing within the initial development phase.

Chapter 4: Proposed Design

4.1 Block Diagram of the proposed system

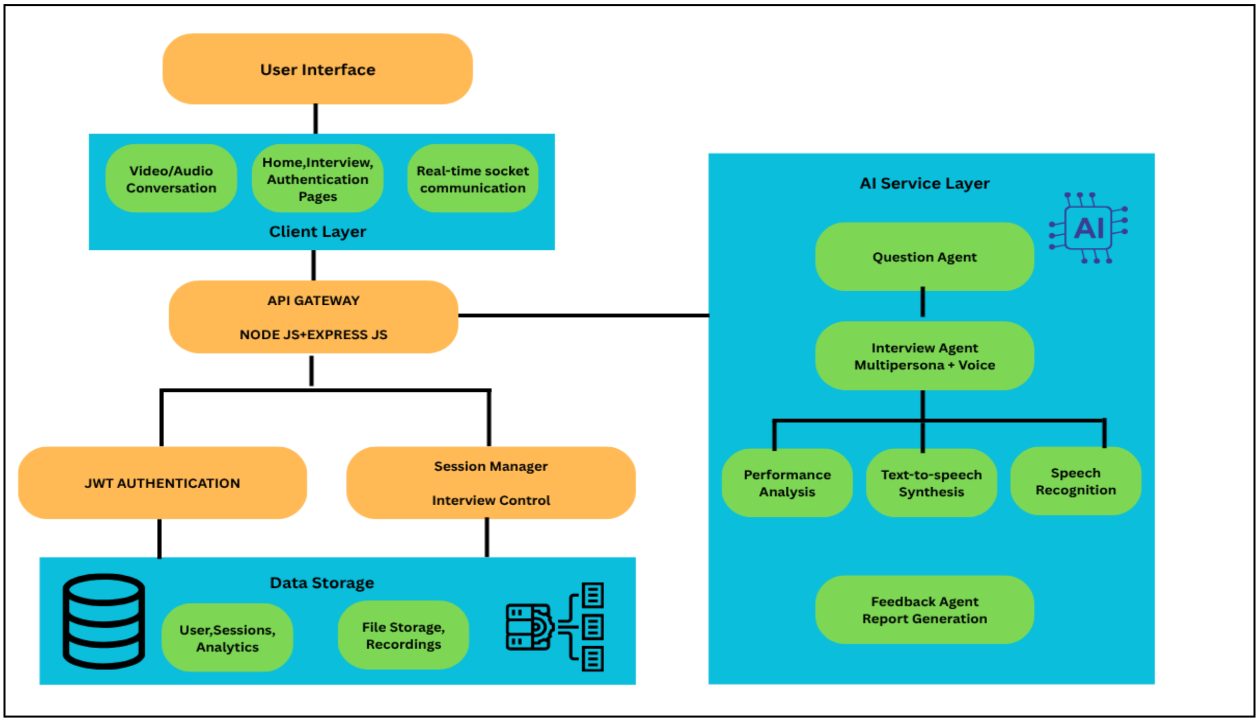

Fig1.Block Diagram

The above block diagram represents the overall system architecture of InterviewIQ, showing how different components interact to deliver AI-driven mock interviews.

- **User Interface Layer:**
  This is the front-end where users interact with the system. It includes pages for home, login, interview sessions, and real-time video/audio conversations. Communication with the backend happens via socket connections for live interaction.
- **Client Layer:**
  Built with **React.js**, this layer handles all visual and interactive elements, manages user input/output, and maintains smooth real-time communication during interviews.
- **API Gateway (Node.js + Express.js):**
  Acts as the bridge between the frontend and backend services. It manages data flow, handles API requests, performs authentication, and connects with the AI service layer.
- **JWT Authentication & Session Manager:**
  Ensures secure login and maintains interview sessions. JWT (JSON Web Token) authentication validates users, while the Session Manager handles interview control and timing.

- **Data Storage:**
  Uses **MongoDB** to store user data, session history, analytics, and recorded files. It ensures persistence of all interview-related data.
- **AI Service Layer:**
  This is the system's intelligence core. It includes multiple AI agents such as:
  - **Question Agent** – Generates adaptive questions using LLMs.
  - **Interview Agent** – Conducts the interview with selected persona and voice.
  - **Speech Recognition Module** – Converts user speech to text.
  - **Text-to-Speech Module** – Converts AI responses to realistic voice.
  - **Performance Analysis & Feedback Agent** – Evaluates speech, gestures, and responses to generate detailed reports and analytics.

Together, these layers ensure real-time, secure, and intelligent interaction between the user and the AI interviewer, providing a realistic and data-driven interview experience.

4.2 Modular diagram

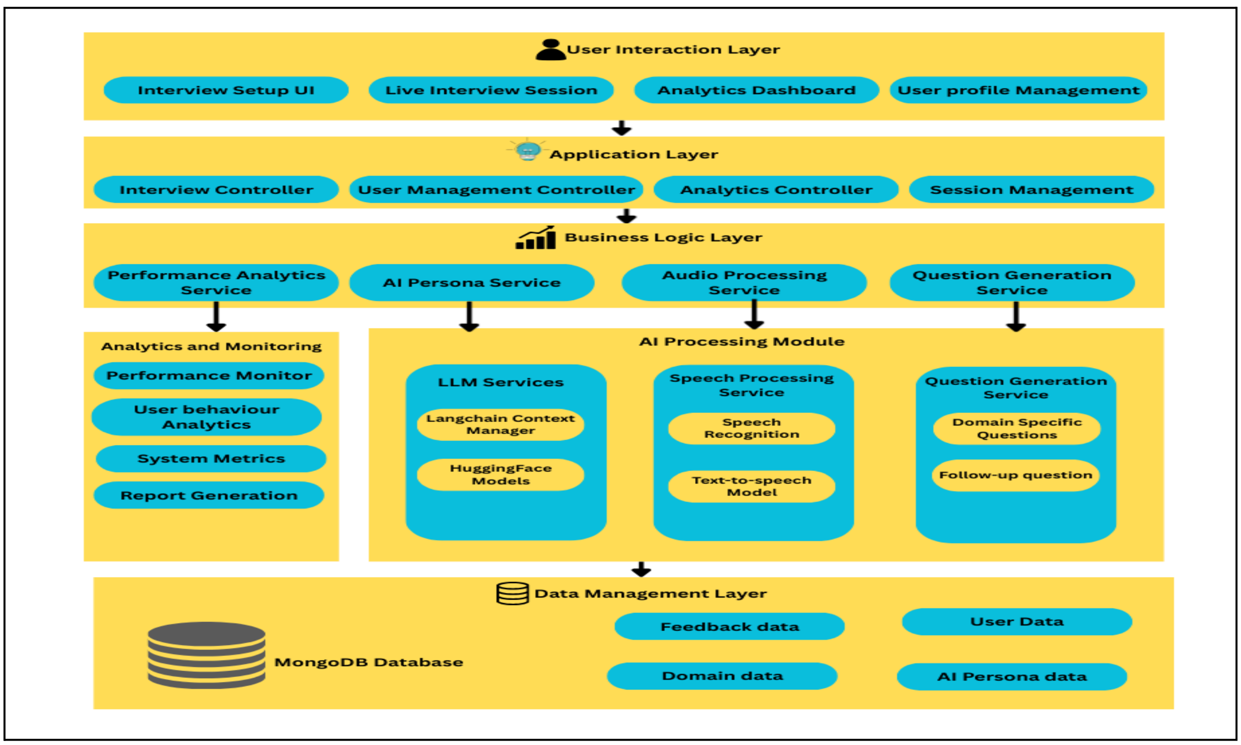

Fig2.Modular Diagram

The modular diagram of InterviewIQ represents the complete workflow of the system and how its major components interact to deliver AI-powered mock interviews.

The User Interaction Layer is the front-end interface where users set up interviews, join live sessions, view analytics, and manage profiles. Below it, the Application Layer controls user management, interview sessions, and analytics processing, ensuring smooth communication between the interface and backend.

The Business Logic Layer handles the system's core operations through modules like Performance Analytics, AI Persona Service, Audio Processing, and Question Generation, which together manage evaluation, interviewer behavior, voice processing, and adaptive questioning.
At the core, the AI Processing Module integrates LLM services, speech processing, and question generation models to conduct realistic, domain-specific, and adaptive interviews. The Analytics and Monitoring unit tracks performance metrics and generates user reports, while the Data Management Layer (MongoDB) securely stores all user data, feedback, and AI configurations.
Overall, this modular structure ensures scalability, intelligence, and real-time adaptability.

4.3 Detailed Design

DFD Diagram - Level 0

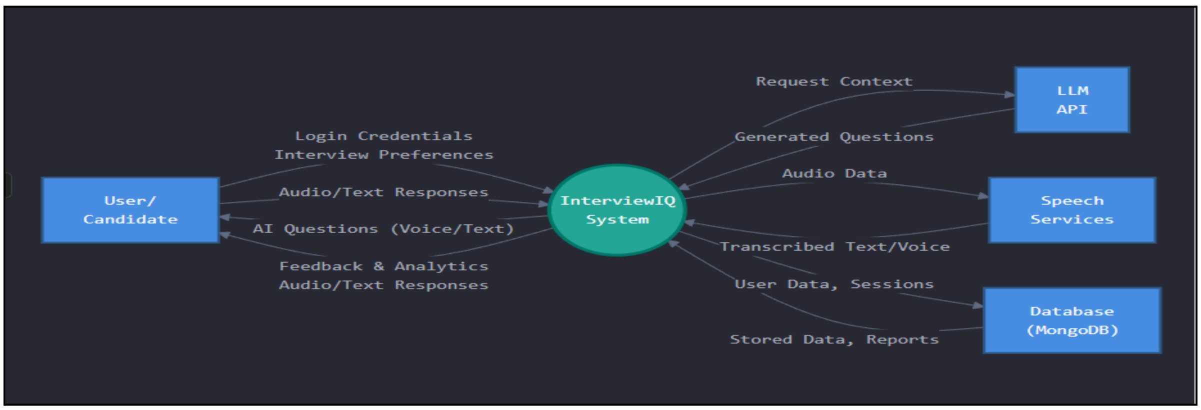

Fig3.DFD Level-0 Diagram

DFD Diagram - Level 1

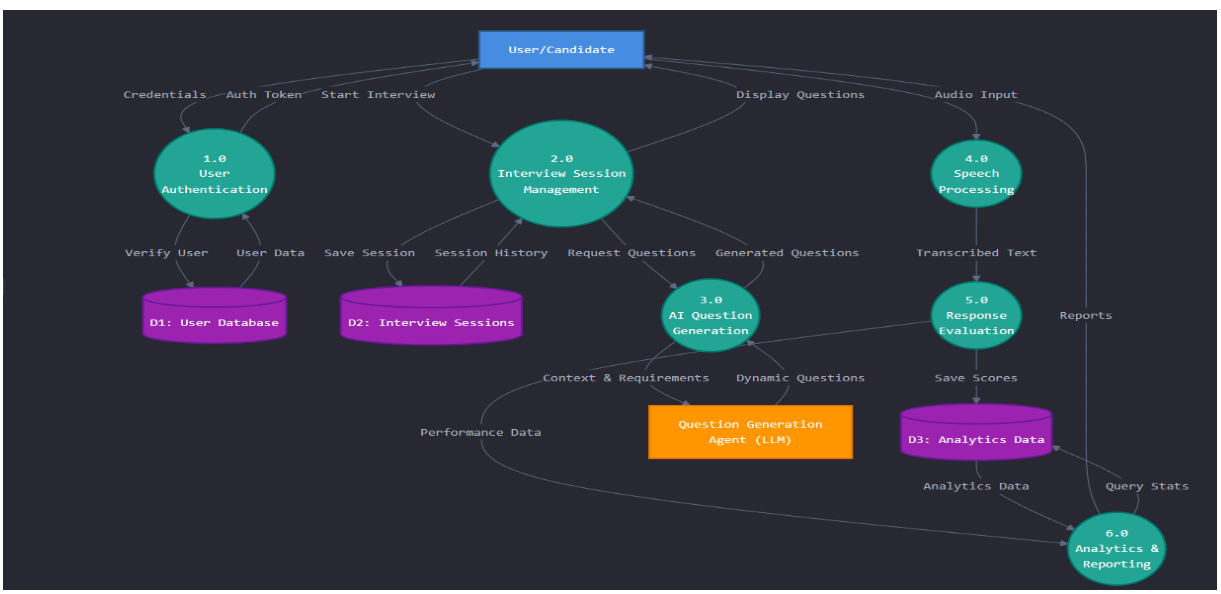

Fig 4.DFD Level-1 Diagram

Flowchart:

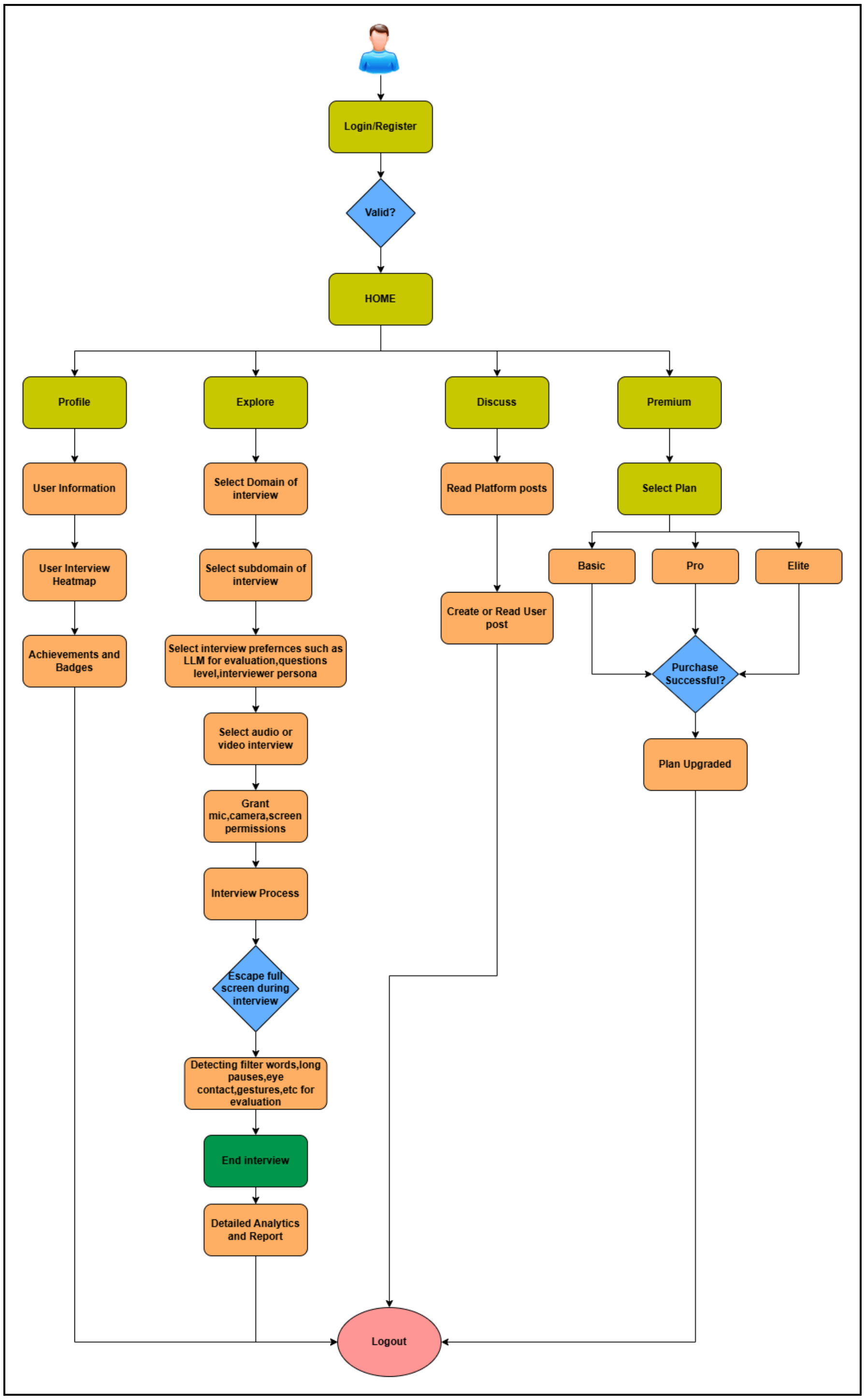

Fig5.Flowchart

4.4. Proposed algorithms

Step No	Process/Description
1	**User Authentication and Initialization:** The user logs into the system, selects the interview mode (audio/video), preferred interviewer persona, and desired domain or subdomain.
2	**Interview Session Setup:** The system initializes the interview session through the API Gateway, establishes real-time socket communication, and activates the selected AI interviewer.
3	**Dynamic Question Generation:** The **LLM-based Question Agent** generates context-aware questions according to the user's chosen field and previous responses. Difficulty level and flow adapt dynamically.
4	**Audio/Video Input Processing:** The system captures the user's voice and video in real time. Audio is processed through **speech-to-text (Whisper ASR)** and analyzed for filler words, pauses, tone, and fluency. Video input is analyzed using **OpenCV and MediaPipe** for gestures, eye contact, and expressions.
5	**Response Evaluation:** The **AI Interview Agent** evaluates both verbal and non-verbal responses, scoring parameters such as confidence, clarity, engagement, and technical accuracy.
6	**Feedback and Report Generation:** The **Feedback Agent** compiles results into a comprehensive analytics report highlighting strengths, weaknesses, and overall performance trends.
7	**Data Storage and Analytics Update:** Session data, reports, and metrics are securely stored in **MongoDB**, updating the user's profile and progress analytics dashboard.

4.5 Project Scheduling & Tracking using Time line / Gantt Chart

The InterviewIQ Timeline 2025 – Gantt Chart illustrates the structured project development plan, highlighting major phases, tasks, and their respective durations from August to November 2025. The project begins with the Planning phase, covering topic discussion, requirement gathering, and synopsis submission. It then progresses to the Initial Working phase, focusing on frontend and backend development. Following the first review, the team worked on feedback implementation, conducted existing system research, and developed the Flask API for AI and audio processing. The Advanced Development Phase involved integrating AI agents, speech-to-text systems, and real-time socket communication, followed by Review-2. Finally, the

Final Touch phase focuses on improving features based on feedback and completing Review-3. The status indicators show that most components are successfully completed or in the final stages of progress, reflecting a well-organized and timely project execution.

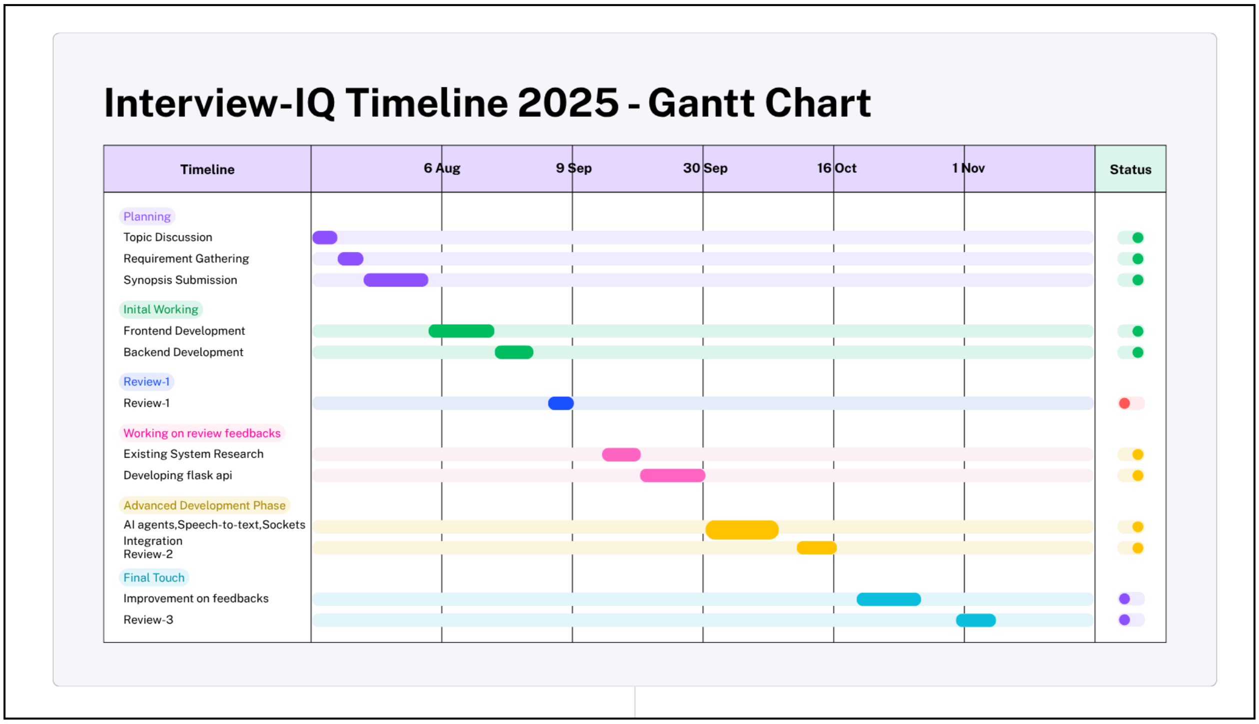

Fig 6.Gantt Chart

5. Results and Discussions

The development of InterviewIQ has successfully reached a major milestone, with most of the system's core functionalities implemented and tested. The frontend module, built using React.js, has been completed with an intuitive and responsive user interface that allows seamless navigation across modules such as interview setup, live session, analytics dashboard, and profile management(Fig6,Fig7). On the backend, major services have been developed using Node.js and Express.js, ensuring secure data flow and efficient API handling. User authentication, session management, and interview preparation modules are fully functional, providing a smooth and secure experience for all users.

A dedicated Flask-based API has been successfully designed and integrated for handling AI operations such as interview question generation using LLMs and speech-to-text synthesis for real-time interaction. The system now allows users to choose their preferred interviewer persona (from four AI interviewers) and even select which LLM they want to evaluate their responses.(Fig.8) Real-time interview sessions are running successfully with Socket.io integration between the React client and Flask backend, enabling continuous question-answer communication without noticeable lag.(Fig.9,Fig11)

In terms of performance, the latency in question generation and AI response has been significantly minimized, leading to smoother interviews. The system also detects long pauses and hesitation patterns in user speech, adjusting the confidence score accordingly. The filler word detection module is currently under refinement to enhance its accuracy and evaluation reliability. After each interview, users receive a comprehensive analytics report that summarizes their strengths, weaknesses, and performance trends. The platform also supports multiple interview domains, including Technology, Finance, Marketing, and Sales, each with several specialized subfields.(Fig.10) The MongoDB database has been successfully configured to store user information, session history, and analytics data, ensuring persistence and scalability.

Overall, the system has achieved excellent progress across both frontend and backend components. With stable real-time communication, AI integration, and dynamic interview evaluation, InterviewIQ has demonstrated strong proof of concept and functionality. The upcoming focus will be on further improving speech evaluation accuracy, filler word detection, and report visualization for enhanced user experience and precision in feedback generation.

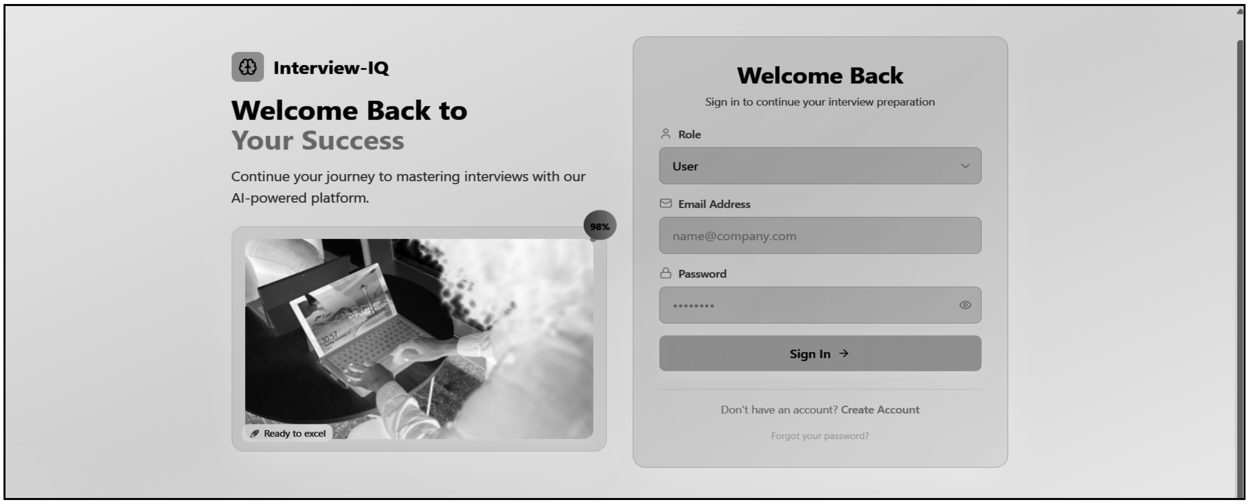

Fig 7.Login Page

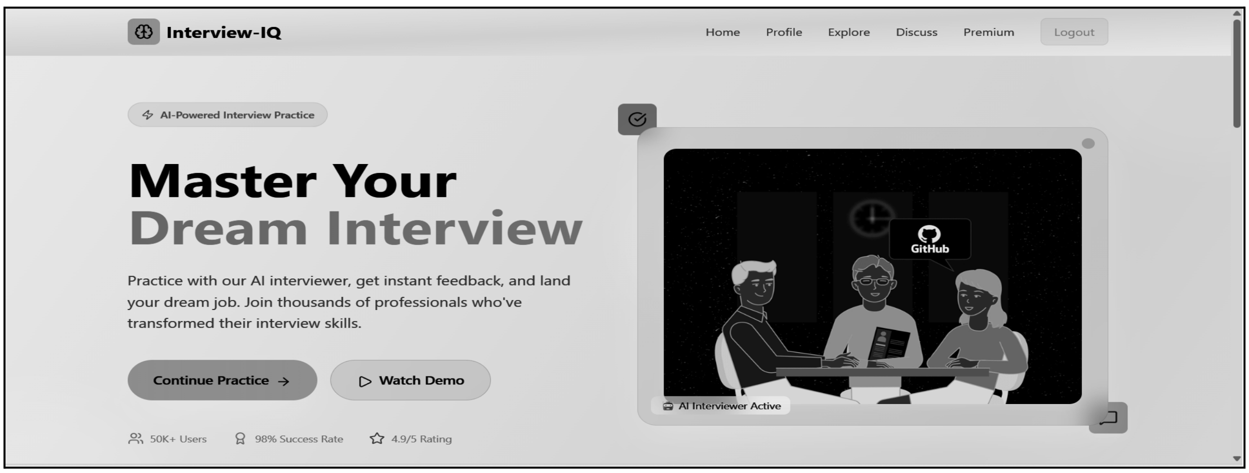

Fig8.Home Page

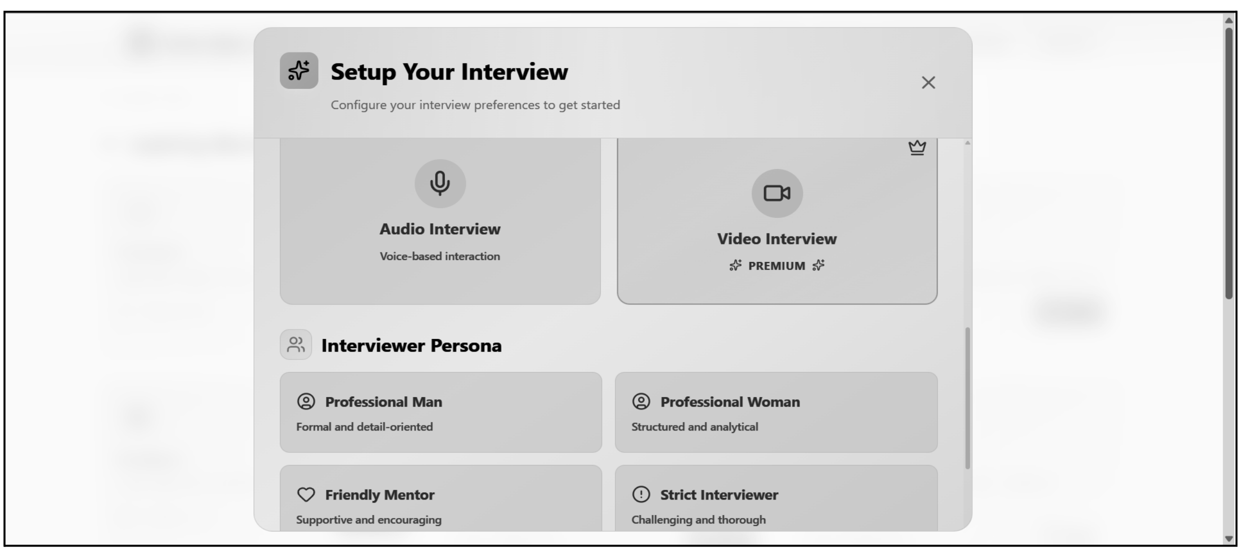

Fig9.Interview Settings Page

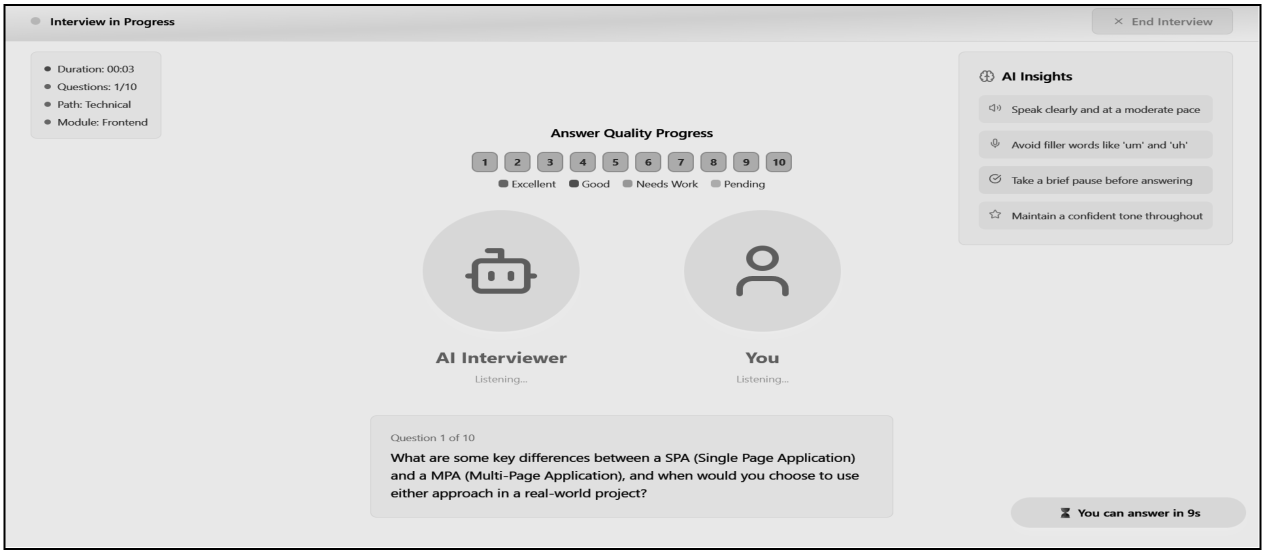

Fig 10.Interview Page

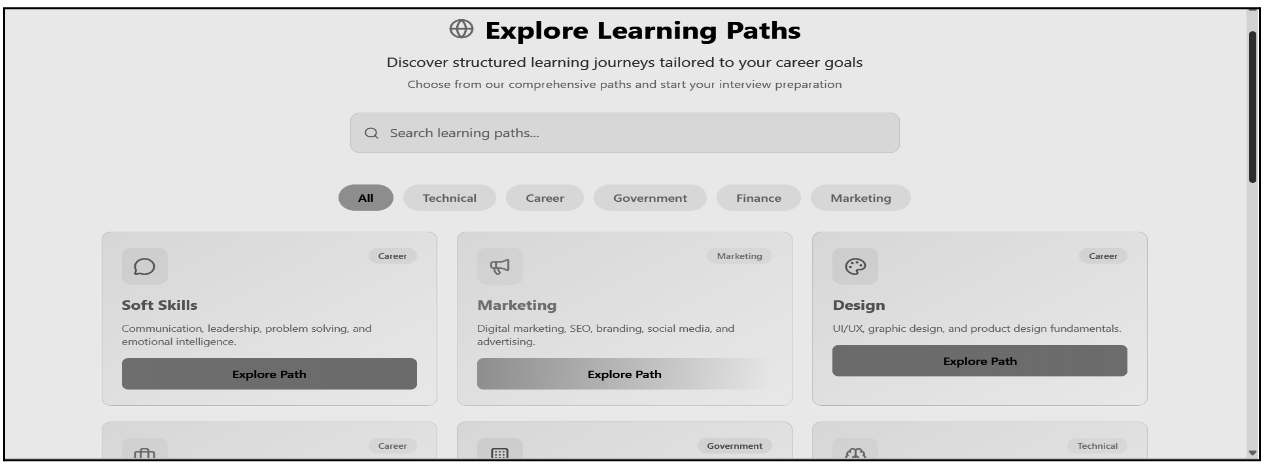

Fig11.Explore Page

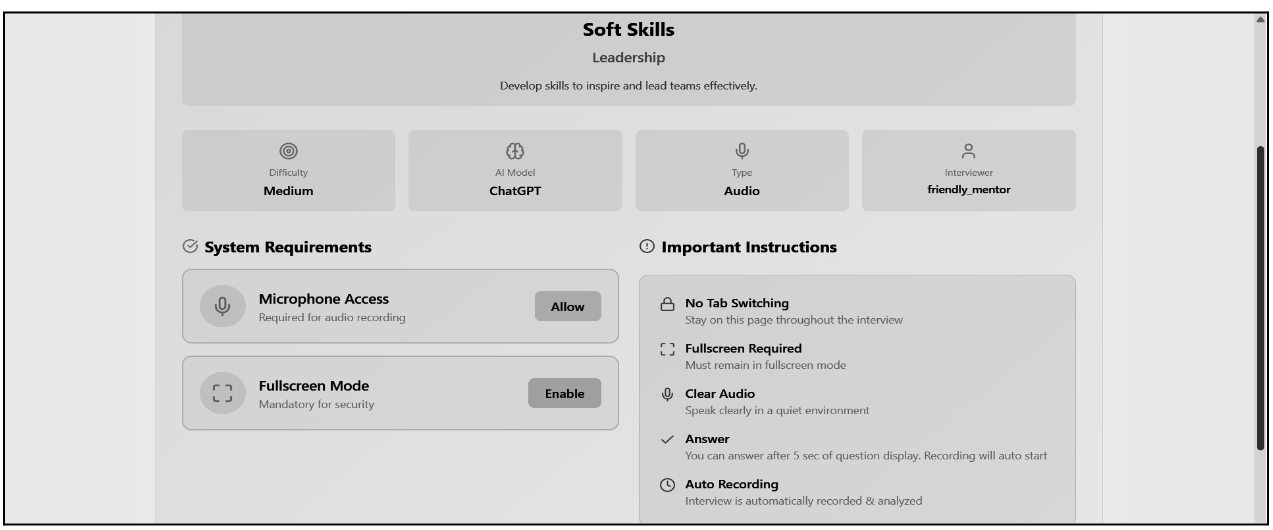

Fig12.Interview Setup Page

6. Plan of action for the next semester

In the upcoming semester, the primary focus will be on enhancing the intelligence, interactivity, and analytical depth of InterviewIQ to make it a more robust and feature-complete system. The foremost goal is to improve the accuracy of filler word detection — particularly for expressions such as “uh,” “umm,” and “ahh.” Advanced speech-processing techniques and refined acoustic modeling will be incorporated to ensure precise identification and scoring impact of such hesitation patterns.

Currently, the system supports only audio-based interviews; in the next semester, the team will implement and complete the video interview module using OpenCV and MediaPipe. This addition will enable facial expression recognition, eye contact tracking, and gesture monitoring, allowing for a comprehensive behavioral evaluation alongside verbal analysis. The analytics and feedback module will also be upgraded to include detailed per-answer walkthroughs, showing filler word counts, response times, and engagement levels. Furthermore, graphical analytics dashboards will be introduced to present insights such as performance trends, communication patterns, and improvement areas using visual charts and metrics.

A key innovation planned for the next phase is the introduction of a Retrieval-Augmented Generation (RAG)-based interview feature, where users can upload their PDFs, books, or notes, and the AI interviewer will generate and evaluate questions directly from those documents. This feature will personalize interview preparation by aligning questions with the user’s own study material — a unique differentiator of InterviewIQ.

Additionally, secondary tasks will include optimizing WebRTC and socket performance to minimize latency during live interactions, ensuring faster response times and smoother AI-user communication. Collectively, these improvements will make InterviewIQ more interactive, accurate, and personalized, delivering an advanced and data-rich mock interview experience.

7. Conclusion

The development of InterviewIQ – An AI-Powered Mock Interview Platform marks a significant step toward modernising and personalising interview preparation through the integration of advanced Artificial Intelligence technologies. The system successfully simulates realistic interview scenarios using LLM-based question generation, real-time audio interaction, and intelligent performance evaluation. By allowing users to choose from multiple interviewer personas, domains, and difficulty levels, InterviewIQ delivers a unique, adaptive, and engaging experience that closely mirrors real-world interview environments. The platform also ensures detailed post-interview analytics, enabling users to understand their strengths, weaknesses, and overall progress, thereby boosting confidence and employability.

The current stage has established a strong foundation for the project, with most frontend, backend, and AI components functional. With planned enhancements such as video interview integration, refined filler word detection, improved analytics visualization, and RAG-based personalized interviews, InterviewIQ is poised to become a comprehensive and intelligent solution for digital interview training. In the long run, it holds the potential to redefine AI-driven learning by making professional skill development more accessible, interactive, and effective for users across different career domains.

8. References

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9. Appendix

a. List of Figures

b. List of tables